

# Junhong Cheng

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## EDUCATION

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### Shanghai Jiao Tong University

2023.9 - 2027.6(expected)

B.S in computer science, **ACM Honors Class**

- Cumulative GPA: **90.0**/100

- Selected courses: Mathematical Logic:**A+**, **100/100**, Principle and Practice of Computer Algorithms:**A+**, **97/100**, Compiler Design:**A**, **94/100**. Computer Architecture, Operating System, Machine Learning System, Model Checking.

## RESEARCH INTEREST

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My research interests focus on improving the performance of different computing architectures, especially **Trusted Execution Environments (TEEs)** and **oblivious architectures** such as **ORAM-based** memory systems. I aim to improve the efficiency while preserving the security. I am highly self-directed in learning and can quickly acquire the theoretical and practical skills needed to explore new research challenges.

## RESEARCH EXPERIENCE

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### Network Security and Privacy Protection(NSEC) Lab, SJTU

2025.6 - 2026.6

Participating in project **Attest the Whole, Verify Incrementally** currently under **minor revision** for CCS2026

a second author under the supervision of Prof. Guoxing Chen

decrease the cold start latency by dynamically loading the enclave pages after startup

### Deans Comp Arch and Security Lounge (DCASL), UCSD

2026.5 - now

Participating in the project **Intel Defense Profiling**

co-worked with Luyi Li, and Hosein Yavarzadeh

under the supervision of Prof. Dean Tullsen

Reverse engineering new Intel defense mechanism and measure its overhead

trying to lower its latency with alternatives.

## PERSONAL PROJECTS

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### Mx-compiler

[Link to Repo](#)

A course project of Compiler Design and Implementation.

It's a compiler of Mx\*(a C-like language for education), written in Java.

- Its performance is similar to -O2.
- M2mReg, ADCE are implemented.
- The registers are allocated with SSA-based Register Allocation.

### Verilog-CPU

[Link to Repo](#)

A course project in Computer Architecture.

It's a toy CPU written in Verilog.

- **Tomasulo algorithm** is implemented.
- Basic instructions in RISC-V are supported.

A course project of Principle and Practice of Computer Algorithms, written in Golang.

Raft is a consensus algorithm for managing a replicated log.

- It allows a collection of machines to work as a coherent group that can survive all all non-Byzantine conditions, including network delays, partitions, and packet loss, duplication, and reordering.

AWARDS AND HONORS

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**2023, 2024 Zhiyuan Honors Scholarship** (2 % in SJTU)

TEACHING ASSISTANT EXPERIENCE

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|---------------------------------------------------------------|-------------|
| <b>Great Ideas in Computer Science, Alei Liang</b>            | 2024 Fall   |
| <b>Principle and Practice of Computer Algorithms, Yong Yu</b> | 2025 Summer |

STUDENT WORK

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|----------------------------------------------------------|-----------------|
| <b>Monitor of ACM Honors Class 2023</b>                  | 2024.9 - 2025.9 |
| <b>Organization Committee Member of ACM Honors Class</b> | 2025.9 - 2026.9 |

SKILLS

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|--------------------|-----------------------------------------------------------------------------|
| <b>Programming</b> | C++, Go, Java, Verilog                                                      |
| <b>Tools</b>       | Git, Markdown, L <sup>A</sup> T <sub>E</sub> X                              |
| <b>Languages</b>   | Chinese(Native speaker), English(Fluent, TOEFL 110/120 (5.5/6.0), S5.0/6.0) |